# KA SHING BILL, KU

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## EDUCATION

**University of Oxford | MEng Master of Engineering** **Oct 2022 – Ongoing**
**Exeter College | College Tutor: Prof Michael Osborne** **Expected graduation date: June 2026**
Master's modules:

- C18 Machine Vision and Robotics
- C19 Deep Learning
- C20 Robust and Distributed Control
- C21 Optimal and Learning-Based Control
- C24 Probability, Systems and Perturbation Methods
- C29 Computing Technology

**Queen Elizabeth School, Hong Kong** **Sep 2016 – Jun 2022**
Achieved the highest score of 5 in all exams attended: Physics C: Mechanics, Physics C: Electricity and Magnetism, Physics 1, Physics 2, Calculus BC, Calculus BC: AB Sub-score, Chemistry in the Advanced Placement (AP) exam.

## EXPERIENCES

**Research Project, University of Oxford**
**Towards more human-like learning for artificial neural network language models** **2025 – Ongoing**

- Supervised by Prof Janet Pierrehumbert
- Develop more human-like language learning capabilities in LLMs by exploring more effective word segmentation and tokenization scheme.
- Explore new approaches to tokenization in LLMs, inspired by lingual context, human speech perception, to enhance efficiency and adaptability
- Design new software and language evaluation tasks, building upon existing BabyLM research and incorporating novel ideas
- Applying StateMorph and a beam search, parsing tokenization scheme, seek to obtain improvements in one of the strong BabyLM models by ensuring that the input units are meaningful

**AI Safety Collab Summer 2025, The European Network for AI Safety**
**AI Alignment track** **2025 – Ongoing**

- Specialized in the AI alignment track to keep advanced AI systems safe, secure and beneficial to society
- Weekly discussions on risks, strategies, interpretability, governance and evaluation of cutting edge AI systems

**AI Safety Evals-Paper Reading Club, BlueDot Impact** **2025 – Ongoing**

- Meets weekly to discuss recent and important papers related to AI evaluations
- Engaged in discussions on papers covering topics including Chain of Thought (CoT), AI Alignment, and faithfulness in LLM reasoning.

**BuzzRobot AI Safety Community** **2025 – Ongoing**

- Engaged in discussions in the community of AI developers, researchers and engineers on Slack about the frontiers of AI research
- Actively participated in talks from researchers at Google DeepMind, OpenAI, Meta, NVIDIA, and other leading labs

**Research Project, University of Oxford**
**Designing Intelligent Music Technology: Smart Backing Tracks** **2024 – 2025**

- An industry-first attempt to unlock real time spontaneous backing track adjustments in live performances through natural language inputs from the Music Director, with the whole system running on-device.

- Supervised by Prof David De Roure and Dr Kevin Page.
- Trained a two-classifier system for Text-to-Intent classification, consisting of a fine-tuned BERT and a Multilayer Perception (MLP) neural network.
- Context-aware by retrieving real time musical context from the DAW
- Adapts to each user's unique style and phrase preference when speaking, through incremental fine-tuning with session logs.
- Achieved a 94% classification accuracy with the primary classifier, and an overall accuracy of 89% for the entire context-aware system.

**LLM Local Fine Tuning and deployment, Personal Research** **2025 - Ongoing**
- Developed empathetic conversation model by fine tuning LLaMA 3.1 8B capable of local inference to provide emotional support while preserving user privacy
- Designed and curated dataset of 130+ empathetic conversations addressing emotional distress scenarios including relationship breakdown and grief
- Configured and executed fine-tuning pipeline using QLoRA with 4-bit quantization, adapter rank of 32, and gradient accumulation to overcome local hardware constraints
- Implemented comprehensive safety framework with crisis detection, appropriate boundaries, and professional resource referrals, ensuring responsible deployment of emotional support AI
- Model inference tested on an M1 Pro mac with ~18 tokens/s

**AI Research Intern, Health Bridge International Medical Group Limited** **July 2024 – Sep 2024**
- Developed an activity recognition algorithm to deploy on AI wearable health devices.
- Created a robust data pre-processing workflow handling 45+ hours of raw accelerometer and gyroscope data, implementing median filtering, z-score normalization and anomaly detection techniques to improve signal and data quality
- Implemented a CNN classifier with 2-Conv, 2-Dense layers for recognition of 7 human activities with 87% accuracy

**Engineering Intern(Remote), Health Bridge International Medical Group Limited** **Oct 2023 – Jun 2024**
- Engineered a RNN model for blood glucose level prediction from user-inputted daily blood glucose data with physiological data (heart rate, physical activity)
- Optimized the model to extract physiological data from Apple HealthKit API
- Provide confidence intervals for estimated future glucose levels, and push notifications of potential hyperglycemia events
- The high glucose alert achieved an F1 Score of 0.80 from an independent testing dataset
- Utilized transfer learning to enhance model personalization and accuracy with limited user-specific data.

**Engineering Intern, Health Bridge International Medical Group Limited** **Jun 2023 – Sep 2023**
- Designed a smart online diagnosis redirection algorithm.
- Fine-tuned an NLP model (BERT base uncased) to categorize incoming patient requests.
- Re-directed patient requests to the best available medical expert.
- Reduced average waiting time by over 20%

**Co-organizer, Google Developer Groups (GDG) Devfest Oxford** **Dec 2024**
- Co-organized the GDG Devfest with Gregory McGann at Exeter College, Oxford
- Coordinated venue logistics and secured appropriate facilities accommodating 100+ technology professionals and students
- Supported day-of-event operations including registration, speaker coordination, and attendee engagement

**AI & Machine Learning in Python, Coursework Project, University of Oxford** **2024**
- Supervised by Dr Yangchen Pan, Dr Jindong Gu, Dr Zhiyao Luo, Dr Kevin Sun
- Trained a regression algorithm to model housing prices in California.

- Trained a classification algorithm using SVMs, achieved over 90% validation accuracy on a dataset identifying breast cancer patient
- Trained neural networks for computer vision using PyTorch, achieved over 85% classification accuracy with the MNIST dataset

**High Performance Computing, Coursework Project, University of Oxford 2024**

- Supervised by Prof Wes Armour
- Wrote algorithms in C and CUDA programming language to perform scientific calculations
- Programmed multi-core architectures using OpenMP
- Programmed many-core architectures (NVIDIA GPUs) using CUDA
- Performed the whole project in a Linux environment
- Developed on cloud supercomputers with V100 GPUs from University of Oxford

**Student Ambassador, Department of Engineering Science, University of Oxford 2024 - Ongoing**

- Represents the Department of Engineering Science of Oxford University at various outreach events throughout the year.
- Leads facility tours that emphasize Oxford's cutting-edge research environment to prospective engineering students.
- Articulates how engineers drive global technological advancement and social impact.

**NVIDIA Deep Learning Institute 2022**

- Implemented and fine-tuned a pre-trained deep learning model to distinguish rotten fruits.
- Achieved over 91% classification accuracy.

**Mechanical CAD, Engineering Coursework Project, University of Oxford 2024**

- Supervised by Ms Belinda Hughes
- Designed a walking mechanism from scratch with SolidWorks
- Manufactured and assembled designed parts through 3D printing and CNC
- Passed all terrains within 30s

**Instrumentation & Control Practical, University of Oxford 2024**

- Supervised by Dr. Jack Umenberger and Junaid Memon.
- Designed a PID controller for a Quanser Aero (dual-rotor two degree-of-freedom aerospace system).
- Performed modelling and simulation of the system through Simulink.
- Tuned the PID controller to control power input to the two rotors to achieve disturbance rejection and reference tracking, achieved all the design goals of zero-steady state error and a fast, smooth and stable dynamic response.

**Communications Practical, University of Oxford 2024**

- Supervised by Dr. Eleanor O'Hara, Prof Dominic O'Brien
- Performed Fast Fourier Transform (FFT) algorithm with an oscilloscope to investigate incoming signals and properties of signal propagation along a coaxial cable

**Electrical Practical, University of Oxford 2023**

- Supervised by Dr. Eleanor O'Hara
- Designed and built a programmable electronic music box with switch controls
- Loaded a piece of C code to the integrated circuit board to achieve programmed playback control

**Mechanical Practical, University of Oxford 2023**

- Supervised by Mr R Scott
- Designed a pin-jointed truss structure withstanding 150kN of load
- Optimized the structure's cost and strength through detailed stress analysis via frame2d on MATLAB

**Teaching Assistant, Chinese University of Hong Kong** **2023**

Assisted the lecturer in administrative work and provided guidance to students on their in-class assignments and exercises, in the "Introduction to Number Theory" and "Solving Algebraic Equations" courses

**Committee, Access Abroad Hong Kong (AAHK)** **2022 – Ongoing**

Working with a team of current Hong Kong students from top UK universities to provide free guidance to underprivileged students applying to top UK universities, aiming to unleash the potential of talented students who might be disadvantaged in their application due to their economic background

**Hong Kong University of Science and Technology, HKAGE: Physics Enhancement Programme for Gifted Students Phase II** **Jan-Feb 2020 & 2021**

Ranked 16(in 2020) and 21(in 2019) in Hong Kong in the ultra-intensive Olympiad Physics training and selection programme that selects candidates representing Team Hong Kong in APhO and IPhO competitions

## SCHOLARSHIPS AND AWARDS

**Williams College Winter Study Programme Scholarship** **Jan 2025**

- Jointly funded by Exeter College, Oxford & Williams College, US
- Awarded to 10 students at Exeter College, Oxford for a 10-day study program at Williams College

**AP Scholar with Distinction (College Board of the United States)** **2021**

Awarded by College Board for score of 5(Top Score) in all six subjects attended

**Course Scholarship, Chinese University of Hong Kong** **2021**

Awarded for excellent performance in "Towards Differential Geometry" course by the Department of Mathematics, Chinese University of Hong Kong

## COURSES & CERTIFICATES

**AI Alignment Course, BlueDot Impact**

**Machine Learning Specialization by Andrew Ng**

- Supervised Learning – Regression & Classification
- Advanced Learning Algorithms
- Unsupervised Learning, Recommenders, Reinforcement Learning

**Deep Learning Specialization by Andrew Ng**

- Neural Networks and Deep Learning
- Improving Deep Neural Networks: Hyperparameter Tuning, Regularization and Optimization
- Structuring Machine Learning Projects
- Convolutional Neural Networks
- Sequence Models

## TECHNICAL EXPERIENCE

**Programming Languages:** Python, C++, MATLAB
**ML Libraries:** PyTorch, Tensorflow, Hugging Face Transformers, Pandas, Scikit-learn, Numpy, MatPlotLib, SciPy

## RESEARCH INTERESTS

Machine learning, deep learning, AI safety, AI alignment, computer vision, reinforcement learning, multi-agent systems, LLMs

## LANGUAGE PROFICIENCY

Proficient in English, Cantonese Chinese(native) and Mandarin Chinese(native)